

STACK KINGS

From reactive averages to proactive trade marketing

unlocking latent demand across 9,000 Western Province outlets.

DATA STORM 7.0 · ROUND 2

SK

Historical Data Blinds You



Constrained Outlets

A busy town-center kade may underperform due to stock or credit constraints not lack of demand.



Maxed-Out Villages

A village shop may already be at capacity. Pouring spend there yields zero incremental return.

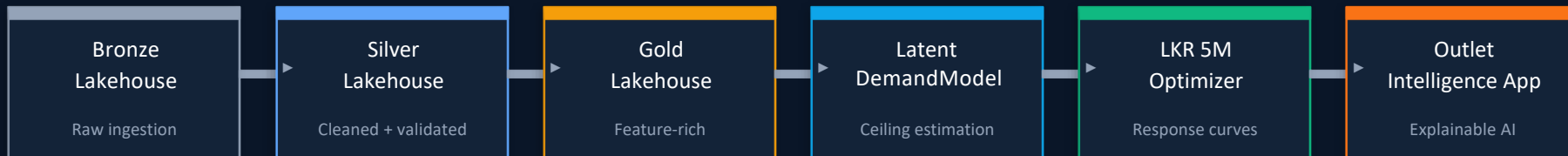


Wasted Trade Spend

Allocating coolers & budgets on past averages leaves latent volume and real revenue on the table.

The Insight: Observed volume is a floor —not a ceiling.

The Potential-Based Allocation Engine



We shift from reactive sales averages → proactive potential-based trade marketing.



Latent Demand Modeling

Treat observed volume as a right-censored lower bound. Ensemble of lookalike ceilings + quantile regression ($\tau=0.90$).



Spatial Intelligence

Exponential distance-decay POI influence. DBSCAN geographic clustering. Non-linear competitive saturation.



Optimizer + Explainability

Diminishing-returns response curves. LKR 5M deployed with +253% lift vs naive allocation. AI-narrated outlet reports.

Unmasking Latent Demand

01 Lookalike Cluster Ceilings
Observed volume treated as the floor not the true demand. True potential lies above.

02 Lookalike Cluster Ceilings
Peer outlets with similar features but no constraints show what's achievable.

03 Quantile Regression ($\tau=0.90$)
Feature-driven upper-tail estimation captures 90th percentile potential per outlet.

04 Ensemble Maximum
Maximum defensible ceiling across all methods per outlet robust and auditable.

RESULTS

1.26x

Median uplift
over historical maximum

100%

Backtest coverage
all outlets validated

37,205

Rejected records
traced with reason codes

327/0

Audit results
PASS / FAIL - full reproducibility

Where You Are Shapes What You Sell

NON-LINEAR



Exponential Distance-Decay POI

Closer footfall drivers (transit, supermarkets, offices) exert greater influence modeled with exponential decay, not flat 3 km counts.

CONTEXTUAL



Competitive Catchment Density

Saturation vs. isolation adjusts each outlet's ceiling. High-density = diminishing returns. Low-density = untapped upside.

INTELLIGENT



DBSCAN Geographic Clustering

Identifies dense commercial clusters vs. rural isolation. Peer group formation is geographically intelligent not just category-based.

Spatial signals are non-linear standard models miss this entirely.

Western Province: LKR 5M

9,000

Western Outlets

LKR 5M

Budget — 100% Utilized

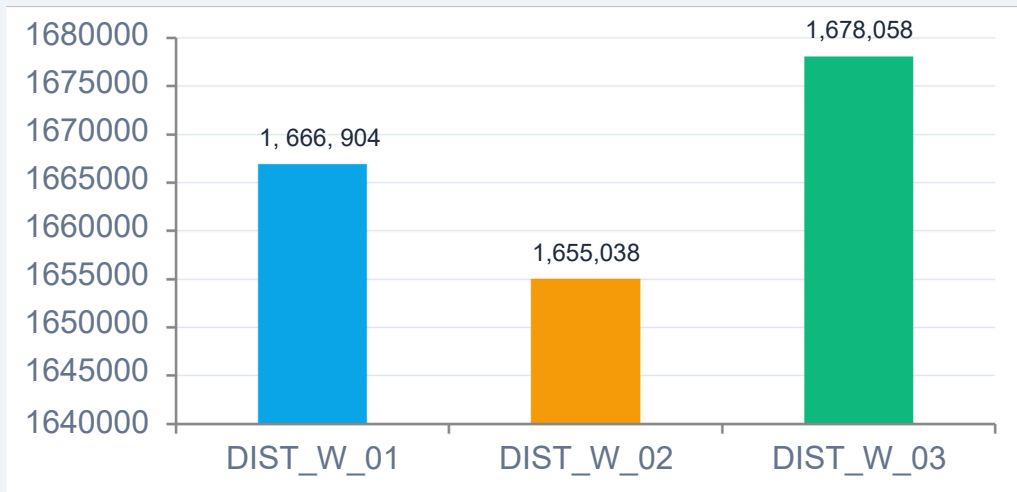
201L

Per LKR 1,000 Spent

+253%

vs Naive Equal Split

Distributor Allocation



HOW THE OPTIMIZER WORKS

- ✓ Diminishing-returns response curves per outlet
- ✓ Spendconcentrates on high-gap, lower-saturation outlets
- ✓ Per-outlet cap to protect field floor coverage
- ✓ Every rupee tied to modeled volume response
- ✓ Fully auditable: optimization_report.csv

Quantified, Auditable Business Impact

~1,000,000 L

Total incremental volume optimized
deployment

Western Province · January 2026

LKR
5.00M

100% utilized — zero waste

201 L
Return

per LKR 1,000 spent

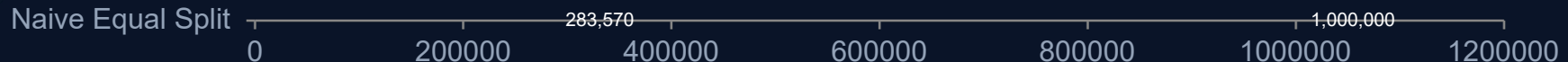
+253%
Lift

vs naive top-500 equal split

100%
Top-100 Gap

outlets receive field floor

Every rupee is tied to a modeled volume response —auditable, defensible, and reproducible.



Outlet Intelligence App



Browse & Filter

20,000 outlets · filter by province, distributor · instant map overview



Explainable AI

"Explain this outlet" → hybrid AI narrative: score, drivers, local conditions, Western spend



Field-Ready

Tablet-optimized for sales reps. No Python required. Works offline with template fallback.



Enterprise-Grade

Medallion architecture. Quarantine store. Full data quality audit trail.

AI STACK

Ollama (Local)

Privacy-first, offline capable

Gemini API

Cloudfallback for richer narratives

Template Fallback

Always works even without AI

DEMO

```
npm run build:clean && npm run start
```


Built for Scale —Starting Western Province



ENTERPRISE ARCHITECTURE

Bronze/Silver/Gold Lakehouse

Quarantine Audit Store

327 PASS / 0 FAIL Audit

Full Reproducibility

Why Stack Kings Wins



End-to-End Solution

Not just a model a full pipeline from raw data to field-ready app with explainable AI.



Enterprise-Grade Architecture

Medallion lakehouse + quarantine store + 100% reproducible audit trail.



Novel Methodology

Right-censored demand modeling + spatial intelligence + ensemble ceiling published-grade thinking.



Fully Defensible Results

+253% over naive baseline. 327 audit checks. Every rupee tied to modeled volume response.

Thank you